ALEX SHEDRACH UCHENNA

Full-Stack Software Engineer | TypeScript / JavaScript | Java / Spring Boot

Founder & Software Engineer, Adcomtechs | Computer Science Undergraduate, University of Nigeria, Nsukka

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GitHub: https://github.com/adcomtechs

Portfolio: https://alexshedrach.vercel.app/

PROFESSIONAL SUMMARY

Full-Stack Software Engineer and Computer Science undergraduate at the **University of Nigeria, Nsukka (UNN)**, with an **ND in Computer Science, Software Engineering option, graduated with Distinction**. I build production-oriented web applications using **TypeScript/JavaScript and Java**, with a strong focus on domain modeling, backend architecture, data modeling, and practical product development.

Founder of **Adcomtechs**, where I design and develop software products around real operational and educational problems in the Nigerian market. My work spans academic technology, education platforms, marketplaces, and business software, with hands-on experience across **Next.js, Node.js, Fastify, PostgreSQL, Prisma, Redis, Redux Toolkit, Tailwind CSS, Docker, Vercel, Railway, Paystack, Flutterwave, Java, and Spring Boot**.

I approach software development from the domain outward: understand the problem, model the business rules, design the system boundaries, then choose the appropriate framework and infrastructure. I am seeking opportunities as a **Full-Stack Software Engineer**, particularly in teams working with **TypeScript/JavaScript or Java/Spring Boot**.

CORE TECHNICAL SKILLS

Languages

TypeScript · JavaScript · Java

Frontend

React · Next.js · Redux Toolkit · Tailwind CSS · HTML · CSS

Backend & APIs

Node.js · Fastify · REST APIs · Spring Boot · Java

Databases & Data

PostgreSQL · Prisma · Redis · IndexedDB · Data Modeling

Architecture & Engineering

Full-Stack Application Architecture · Domain Modeling · Modular Monoliths · Monorepos · Offline-First Architecture · Business Logic Design · API Design · Technical Documentation

DevOps & Deployment

Git · GitHub · Docker · Vercel · Railway · Netlify

Payments & Integrations

Paystack · Flutterwave

Engineering Practices

Framework-independent prototyping · Architecture documentation · Requirements analysis · Product-oriented development · Reusable component architecture · Environment-aware system design

PROFESSIONAL EXPERIENCE

FOUNDER & SOFTWARE ENGINEER

Adcomtechs — Nigeria

2018 – PRESENT

Founded and lead **Adcomtechs**, a hybrid education and software brand evolving into a software company focused on building practical digital products for Nigerian users, students, institutions, professionals, and businesses.

- Design and develop software products from initial problem definition through domain modeling, architecture, implementation, interface development, and deployment preparation.
- Lead technical decisions across both TypeScript/JavaScript and Java/Spring Boot stacks.
- Establish domain-first development practices in which business rules are modeled and validated before framework-specific abstractions are introduced.
- Design products around Nigerian operational realities, including local payment infrastructure, academic workflows, connectivity limitations, and institutional processes.
- Maintain architectural and technical documentation to preserve the reasoning behind major implementation decisions.
- Build reusable patterns and application structures across multiple products rather than treating each application as an isolated project.
- Develop and evaluate systems across education technology, academic marketplaces, retail software, financial tools, and institutional management.

Selected Products Developed at Adcomtechs

AcadClue · GradeMint · Adcomtechs Platform

SELECTED SOFTWARE PROJECTS

ACADCLUE: Nigerian-First Academic Materials Marketplace

Next.js · TypeScript / JavaScript · Node.js · Fastify · PostgreSQL · Prisma · Redis · Redux Toolkit · Tailwind CSS · Docker · Vercel

Live: https://acadclue.vercel.app/

A Nigerian-first academic materials marketplace designed for students, contributors, and eventually academic writers.

- Designed a marketplace model for discovering, uploading, purchasing, and accessing academic resources including past questions, lecture notes, summaries, study guides, project references, and school-specific resources.
- Developed the application as a modular product rather than a collection of isolated frontend screens.
- Worked across frontend, backend, database, marketplace, payments, contributor earnings, and platform architecture.
- Designed backend domains around materials, sellers/contributors, storefronts, carts, checkout, orders, entitlements, wallet balances, earnings, withdrawals, refunds, reviews, notifications, and search.
- Integrated architectural considerations for Paystack and Flutterwave payment workflows.
- Designed the application as a Turborepo modular monolith, keeping clear module boundaries while avoiding premature service extraction.
- Implemented and structured systems for marketplace search, semantic search, academic document similarity, contributor workflows, and financial ledger concepts.
- Designed the application specifically for the Nigerian academic ecosystem rather than adapting a generic international marketplace model.

Status: Active development

GRADEMINT: Framework-Free Grade & Academic Calculation Application

Modern JavaScript · HTML · CSS · Client-Side Storage · Netlify

Live: https://grademint.netlify.app/

A high-performance, framework-free grade calculator designed to provide customizable grading systems and immediate academic calculations.

- Built the application using modern JavaScript without a frontend framework.
- Implemented customizable grading systems and real-time calculation logic.
- Added persistent client-side storage to preserve user configurations and data between sessions.
- Focused on fast execution, minimal dependencies, and clear separation between domain logic and presentation.
- Used the project as a practical demonstration of framework-independent software design and core JavaScript proficiency.

Status: Hosted and publicly accessible

ALEX SHEDRACH PORTFOLIO: Personal Engineering Portfolio

Next.js · React · JavaScript / TypeScript · Tailwind CSS · Vercel

Live: https://alexshedrach.vercel.app/

Personal portfolio website designed, engineered, and deployed independently from start to finish.

- Designed the entire product experience, structure, visual direction, content architecture, and implementation.
- Built reusable components rather than relying on a static template.
- Structured the portfolio around engineering philosophy, selected projects, technical work, and the Adcomtechs story.

- Applied a product-oriented development process covering planning, interface design, implementation, refinement, and deployment.
- Deployed and maintain the application on Vercel.

Status: Live

ADCOMTECHS PLATFORM: Hybrid Physical-Digital Education Platform

Project in active development

Adcomtechs is the broader company platform behind the education and software business.

A hybrid educational platform serving the **Nsukka / Enugu axis of Nigeria**, combining physical learning-center access with online learning and professional service delivery.

- Designed a subscription-based model for students accessing physical education-center facilities and online learning.
- Designed a marketplace-style service layer where professionals and businesses can book services delivered by programme graduates.
- Developed the product concept around the intersection of education, skills development, professional services, and local business needs.
- Building the platform with a strong focus on modular architecture and maintainable system boundaries.
- Applying **Java/Spring Boot** to the backend as part of a deliberate transition into enterprise Java development.

Status: Under development

Repository: Available on GitHub

ENGINEERING APPROACH: Domain First. Framework Second.

I approach software development from the problem domain outward.

Before building framework-specific controllers, components, or database structures, I first work through the underlying business rules and edge cases in a form that can be understood independently of a framework.

This approach has influenced projects such as GradeMint and the broader academic-product work at Adcomtechs. The objective is to understand the problem deeply enough that the eventual framework implementation becomes an expression of the domain rather than the domain being forced into the framework.

Software Should Reflect Its Environment

I design systems around the environment in which they will actually operate.

For Nigerian products, that may mean accounting for local payment infrastructure, academic standards, connectivity constraints, institutional workflows, or operational realities that are often absent from generic software examples.

I consider these constraints part of the architecture rather than exceptions to it.

Documentation Is an Engineering Tool

I document meaningful architectural choices and technical trade-offs so that a system retains its reasoning, not just its implementation.

The objective is to make software easier to maintain, extend, review, and eventually hand over to another engineer.

EDUCATION

UNIVERSITY OF NIGERIA, NSUKKA (UNN)

Bachelor of Science — Computer Science

Current undergraduate student | Year 2 | Expected Graduation: 2028

Currently pursuing an undergraduate degree in Computer Science with a continued focus on software engineering, application development, and system design.

NATIONAL DIPLOMA (ND)

AKANU IBIAM FEDERAL POLYTECHNIC, UNWANA

COMPUTER SCIENCE

Software Engineering Option

Distinction

Completed a National Diploma in Computer Science with a **Distinction**, specializing in the Software Engineering option.

The ND provided the foundation from which I transitioned into broader full-stack and software engineering development.

CERTIFICATIONS & TRAINING

EMFIAS

Data Processing & Desktop Publishing

Certificate in **Data Processing and Desktop Publishing**, completed before transitioning into software development.

This early technical foundation provided practical exposure to computer-based productivity, document processing, and digital workflows before progressing into programming and software engineering.

ADDITIONAL TECHNICAL EXPERIENCE

Full-Stack Development

Experience developing complete web applications across frontend interfaces, APIs, backend services, databases, authentication/business workflows, and deployment.

Backend Development

Hands-on development with **Node.js/Fastify** and growing experience with **Java/Spring Boot**, including API and service-oriented application architecture.

Database Engineering

Experience modeling application data using **PostgreSQL and Prisma**, with Redis used for supporting application infrastructure and performance-oriented workflows.

Modern Frontend Engineering

Experience building application interfaces with **React and Next.js**, using reusable components, state management, responsive design, and utility-based styling.

Application Architecture

Experience designing **modular monoliths, monorepos, domain-oriented modules, and offline-first systems**, with a preference for keeping architecture proportional to demonstrated product requirements.

Deployment

Hands-on deployment and project hosting using **Vercel, Netlify, and Railway**, alongside Git/GitHub-based development workflows and Docker.

PROFESSIONAL INTERESTS

Primary: Full-Stack Software Engineering

Preferred Backend Ecosystems: Node.js / Fastify · Java / Spring Boot

Preferred Languages: TypeScript / JavaScript · Java

Engineering Interests: Backend architecture · distributed application design · database systems · product engineering · developer tooling · enterprise software · education technology

SELECTED PROJECT LINKS

Portfolio
https://alexshedrach.vercel.app/

GitHub
https://github.com/adcomtechs

AcadClue

https://acadclue.vercel.app/

GradeMint

https://grademint.netlify.app/

CAREER OBJECTIVE

To contribute as a **Full-Stack Software Engineer** in an engineering team where I can build reliable software, deepen my expertise in **TypeScript/JavaScript and Java/Spring Boot**, and grow through exposure to real production systems, experienced engineers, code review, testing, system design, and collaborative product development.

I am particularly interested in teams that value strong engineering fundamentals, thoughtful architecture, maintainable code, and software that solves real user problems.

REFERENCES

Available upon request.